

ECE448 Spring 2025

Lesson 03

Signal Space



UNITED STATES
AIR FORCE
ACADEMY

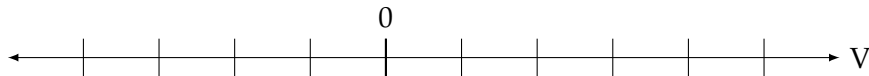
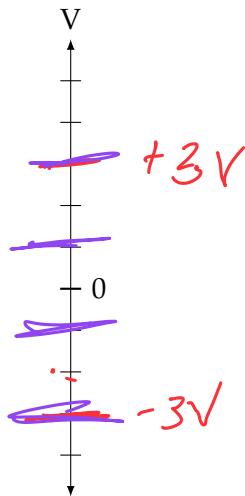
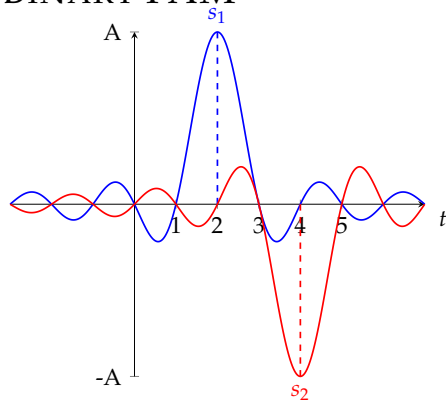
NOMINAL SCHEDULE

Lesson	Topics	Notes
1	Course Overview & Introduction	
2	Noise Figure & Link Budget	
3	Signal Space	
4	PAM Theory	
5	ASCII/PAM	Theory Project Due
6	ASCII/PAM	
7	ASCII/PAM	
8	GRC: OOT Modules	
9	GRC: Custom Blocks	PAM Project Due
10	GRC: Custom Blocks	
11	CW	
12	CW	Custom Block Project due
13	CW	
14	DTMF	
15	DTMF	CW Project Due
16	Flex Day	Final Project Intro
17	FHSS - Tx	
18	FHSS - Rx	DTMF Project Due
19	Midterm (in-class portion)	
20	Flex day	

LESSON 3 LEARNING OBJECTIVES

- 1.1 Understand and articulate the difference between mFSK, mASK, PAM, and mPSK signals.
- 1.2 Calculate the probability of error for each.
- 1.3 Draw the constellation diagram for a given modulation technique and

BINARY PAM

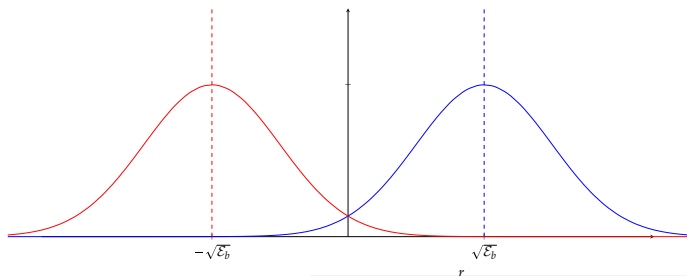


PAM PROBABILITY OF ERROR P_b

- Assume the energy per bit in a PAM pulse is _____.
- In AWGN channel, we have the received signal as: $r = s_k + n$
- The conditional probability of receiving each pulse:

$$p(r|s_1) = \frac{1}{\sqrt{\pi N_0}} e^{-(r - \sqrt{\mathcal{E}_b})^2 / N_0}$$

$$p(r|s_2) = \frac{1}{\sqrt{\pi N_0}} e^{-(r + \sqrt{\mathcal{E}_b})^2 / N_0}$$



PAM P_b

Given a decision threshold of 0, the probability of an error for s_1 tx'd is:

$$P(e|s_1) = \int_{-\infty}^0 p(r|s_1)dr = \frac{1}{\sqrt{\pi N_0}} \int_{-\infty}^0 e^{-(r-\sqrt{\mathcal{E}_b})^2/N_0} dr$$

PAM P_b

Given a decision threshold of 0, the probability of an error for s_1 tx'd is:

$$\begin{aligned} P(e|s_1) &= \int_{-\infty}^0 p(r|s_1)dr = \frac{1}{\sqrt{\pi N_0}} \int_{-\infty}^0 e^{-(r-\sqrt{\mathcal{E}_b})^2/N_0} dr \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{-(r-\sqrt{\mathcal{E}_b})^2/N_0} dr = Q\left(\sqrt{\frac{2\mathcal{E}_b}{N_0}}\right) \end{aligned}$$

PAM P_b

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Since both signals are equally likely...

PAM Probability of Error

$$P_b = \frac{1}{2}P(e|s_1) + \frac{1}{2}P(e|s_2) = Q\left(\sqrt{\frac{2\mathcal{E}_b}{N_0}}\right)$$

PAM P_b

The energy “distance” between the two signals is $d_{12} = 2\sqrt{\mathcal{E}_b}$,
implying:

$$\mathcal{E}_b = \frac{d_{12}^2}{4}$$

PAM P_b

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implying:

$$\mathcal{E}_b = \frac{d_{12}^2}{4}$$

Leading to the big result....

PAM Probability of Error

$$P_b = Q\left(\sqrt{\frac{d_{12}^2}{2N_0}}\right)$$

ORTHOGONAL P_b

PAM signals are antipodal, but what if the signals are orthogonal?

$$s_1 = (\sqrt{\mathcal{E}_b}, 0)$$
$$s_2 = (0, \sqrt{\mathcal{E}_b})$$

Orthogonal Signals P_b

$$P_b = Q \left(\sqrt{\frac{\mathcal{E}_b}{N_0}} \right)$$

m -ARY PAM P_b

m -ary PAM signals

$$s_m = \sqrt{\mathcal{E}_g} A_m$$

m -ary PAM

$$P_m = \frac{2(M-1)}{M} Q \left(\sqrt{\frac{6 \ln(M) \mathcal{E}_{b,\text{avg}}}{(M^2-1) N_0}} \right)$$

m -ARY PAM P_b (BAND-LIMITED)

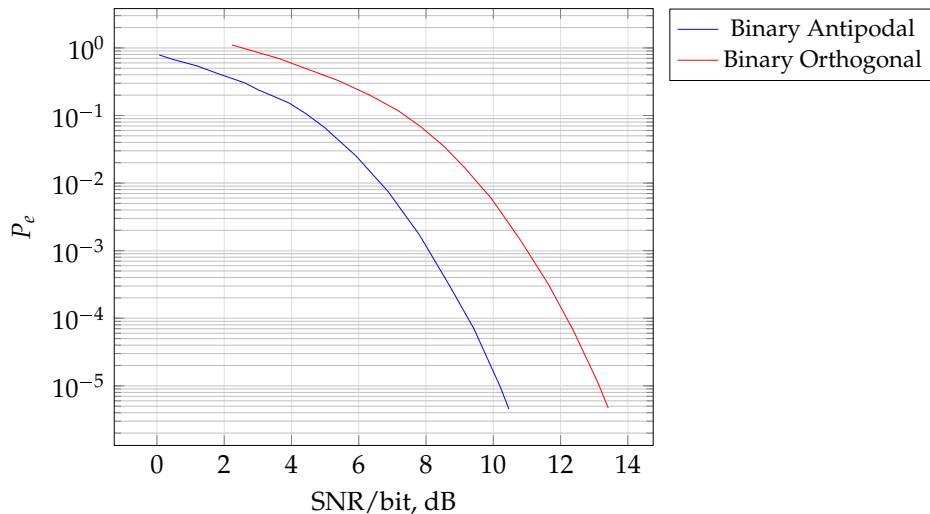
m -ary PAM signals

$$s_m = \sqrt{\mathcal{E}_g} A_m$$

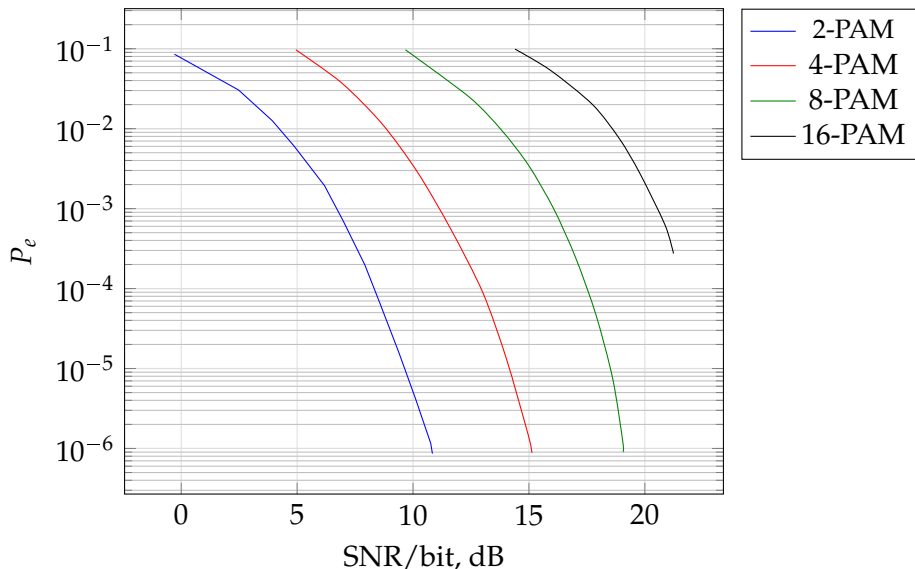
m -ary PAM

$$P_m = \frac{2(M-1)}{M} Q \left(\sqrt{\frac{6\mathcal{E}_{b,\text{avg}}}{(M^2-1)N_0}} \right)$$

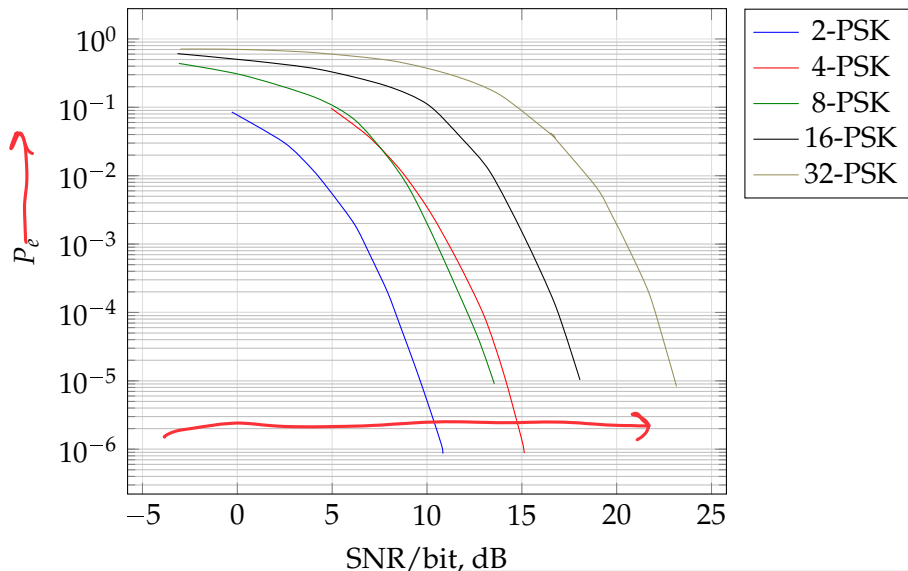
BINARY SIGNALS



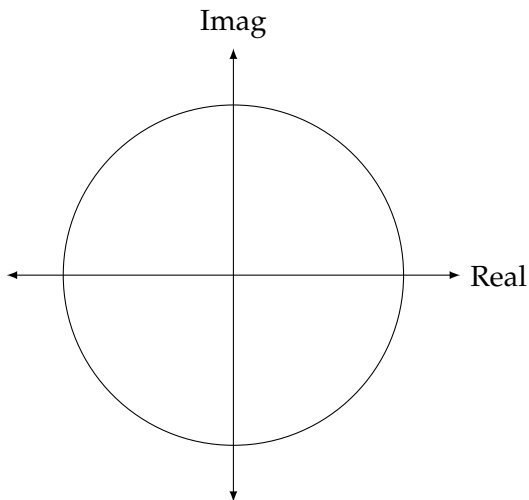
PAM SIGNALS



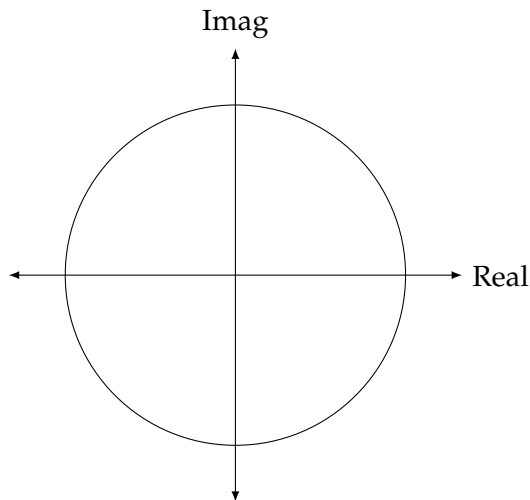
PSK SIGNALS



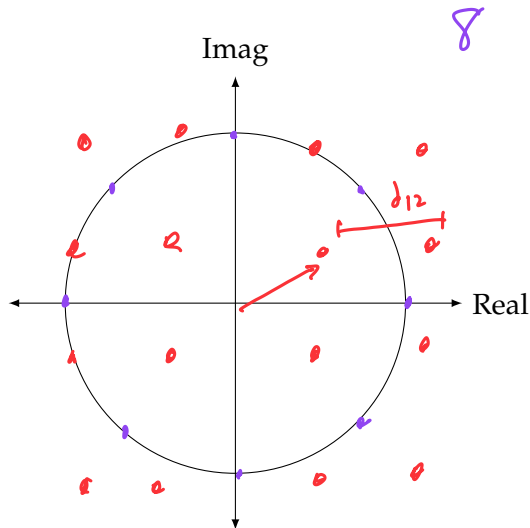
CONSTELLATION DIAGRAM: 2-PAM



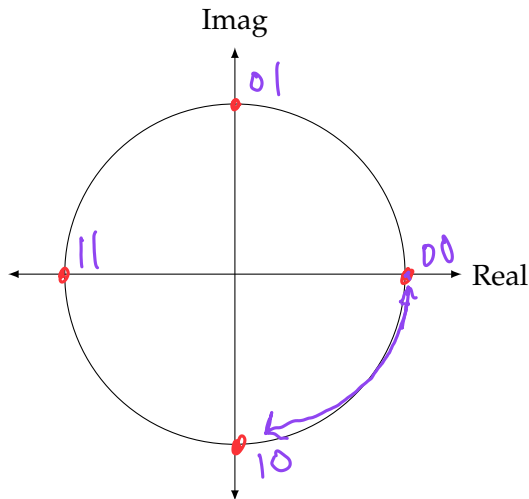
CONSTELLATION DIAGRAM: 4 PAM



CONSTELLATION DIAGRAM: M-PSK



GRAY CODING



Gray Coding

Each successive symbol varies only by 1 bit.

MAP OUT GRAY CODE FOR 8-PSK

